Ch. 5 CW Pt. 2 – Solns

Principles of Microeconomics

September 21, 2025

1. (a) Increase – the introduction of additional substitutes will increase the substitutability of Ford SUVs. That should increase the price elasticity of demand.

   (b) Decrease – the removal of substitutes will decrease the substitutability of Ford SUVs. That should decrease the price elasticity of demand.

   (c) Increase – if people think that SUVs are uniquely safe, that will decrease their substitutability, so the price elasticity of demand should increase.

   (d) Increase – a longer time horizon gives people more time to find substitutes and companies more time to introduce substitutes, such as the four-wheel-drive cargo van. That should increase the price elasticity of demand.

2. (a)

$$\eta = \left| \frac{\%\Delta Q_D}{\%\Delta P} \right|$$

$$\%\Delta Q = \frac{6-5}{\frac{6+5}{2}} = \frac{2}{11}$$

$$\%\Delta P = \frac{4-5}{\frac{4+5}{2}} = \frac{-2}{9}$$

$$\eta = \left| \frac{\frac{2}{11}}{\frac{-2}{9}} \right| = \frac{9}{11}$$

   (b) $\eta < 1$, so demand for gas is inelastic.

   (c) Demand would be more elastic. A longer time horizon gives people more time to find substitutes or change their behavior so that they consume a different amount of gas. As a result, gas becomes more substitutable, and the price elasticity of demand increases.

3. (a)

$$PES = \frac{\%\Delta Q_S}{\%\Delta P}$$
$$1.6 = \frac{\%\Delta Q_S}{-0.18}$$
$$-0.288 = \%\Delta Q_S$$

Q_S decreases by 28.8%.

(b)

$$IED = \frac{\%\Delta Q_D}{\%\Delta I}$$
$$0.6 = \frac{\%\Delta Q_D}{\frac{42{,}000-38{,}000}{40{,}000}}$$
$$0.6 = \frac{\%\Delta Q_D}{0.1}$$
$$0.06 = \%\Delta Q_D$$

Q_D increases by 6%.

(c) The cross-price elasticity of demand would be negative because the two goods are complements. An increase in the price of iPhones will decrease the quantity demanded of portable chargers.

(d) Let $x = \%\Delta Q_D$.

$$TR = PQ$$
$$1.02TR = 1.05P * (1+x)Q$$
$$1.02TR = 1.05(1+x)PQ$$
$$1.02 = 1.05(1+x)$$
$$\frac{1.02}{1.05} - 1 = x$$
$$-0.03 = x$$
$$\eta = \left|\frac{-3\%}{5\%}\right| = 0.6$$

4. The technological breakthrough causes supply to shift right. As a result, price decreases and quantity increases. For total revenue to go up, the increase in quantity must be proportionately larger than the decrease in price. That will be true if demand is elastic.

